

★ WEEK 15 · PART 1 · PRACTICE

When a Confident Number Costs You

Two Cases, Worked by Hand

A wandering beta and a lucky Sharpe - computed step by step

Part 1 in the real world, with the arithmetic shown. We compute a beta and a Sharpe by hand from tiny datasets, watch each point estimate mislead, and update our belief into an honest band - one line at a time.

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PART 1 · APPLICATIONS

Two Cases, Computed by Hand

Case A: compute a stock's beta from five days, see it move, then update it into a band. Case B: compute a Sharpe from six months and find its interval spans zero. Every step is shown; then a Claude Code lab.

What Part 1 Told Us

Now we make it concrete - with numbers

1

A point estimate hides its uncertainty

A single fitted number looks precise but omits how much it could be wrong - especially the parameter uncertainty a filter ignores.

2

Honest doubt has two layers

The quantity itself moves, and we don't know the model's dials. A truthful band includes both, and we will see it emerge from the arithmetic.

3

A posterior turns a line into a fan

A Bayesian update replaces the confident line with a band - and it is just a weighted average you can compute by hand.

Everything here is done with a calculator: five days of data, six months of returns, one update rule.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a tiny, real-looking dataset.
- Compute the point estimate by hand, step by step.
- Then find its honest band - also by hand.

What you should watch

- How the single number looks confident.
- How wide the honest band actually is.
- How the band would change the decision.

The Wandering Market Beta

Every hedge and cost-of-capital rests on it

A stock's beta - its sensitivity to the market - sets the hedge you put on and the discount rate you value it with. It is usually reported as one number. We will compute that number from five days of data, then watch it move and learn to put a band on it.

Where beta is used

- Hedging: how much index to short.
- Cost of capital: the discount rate.
- Risk: the stock's market exposure.

What we'll do by hand

- Estimate beta from five days.
- See two windows disagree.
- Update belief into a band.

Beta as a Drifting Hidden State

The dynamic CAPM, in state-space form

$$\begin{aligned}\text{return: } r_{i,t} &= \alpha + \beta_t r_{m,t} + \varepsilon_t \\ \text{drift: } \beta_t &= \beta_{t-1} + \eta_t\end{aligned}$$

The two equations

- Return: stock equals alpha plus beta times the market, plus noise.
- Drift: beta wanders as a random walk.
- Beta is the hidden state we infer.

Why state-space

- It lets beta move, not stay fixed.
- It yields a band, not just a line.
- The Part 1 model, applied to CAPM.

The Data: Five Days

Compute this stock's beta by hand

Day	Market r_m (%)	Stock r_i (%)
1	+1.0	+1.4
2	-0.5	-0.3
3	+2.0	+2.6
4	-1.0	-1.0
5	+0.5	+0.8

Goal: estimate $\beta = \text{Cov}(\text{market}, \text{stock}) / \text{Var}(\text{market})$. We do it in three lines on the next slide.

Beta = Cov / Var

Three lines with a calculator

$$\begin{aligned}\bar{r}_m &= \frac{1.0-0.5+2.0-1.0+0.5}{5} = 0.40, & \bar{r}_i &= \frac{1.4-0.3+2.6-1.0+0.8}{5} = 0.70 \\ \text{Cov} &= \frac{1}{5} \sum (r_m - \bar{r}_m)(r_i - \bar{r}_i) = \frac{6.75}{5} = 1.35 \\ \text{Var} &= \frac{1}{5} \sum (r_m - \bar{r}_m)^2 = \frac{5.70}{5} = 1.14 \\ \hat{\beta} &= \frac{\text{Cov}}{\text{Var}} = \frac{1.35}{1.14} = \mathbf{1.18}\end{aligned}$$

The single best estimate is beta = 1.18. It looks exact - but it came from just five noisy days.

Two Windows, Two Betas

The number moves as the window moves

Slide the window forward

- Days 1-5 gave beta = 1.18.
- The next five days might give, say, 0.90.
- Same stock, a very different number.

What this tells you

- The 'one beta' is really a moving target.
- Each window is a noisy snapshot.
- No single number is the honest answer.

The Rolling-Regression Beta

Repeating that five-day calculation on a sliding window gives one jumpy line: it moves around a lot yet reports no uncertainty. Is a jump real signal, or just window noise? The line cannot say.



Update the Belief Into a Band

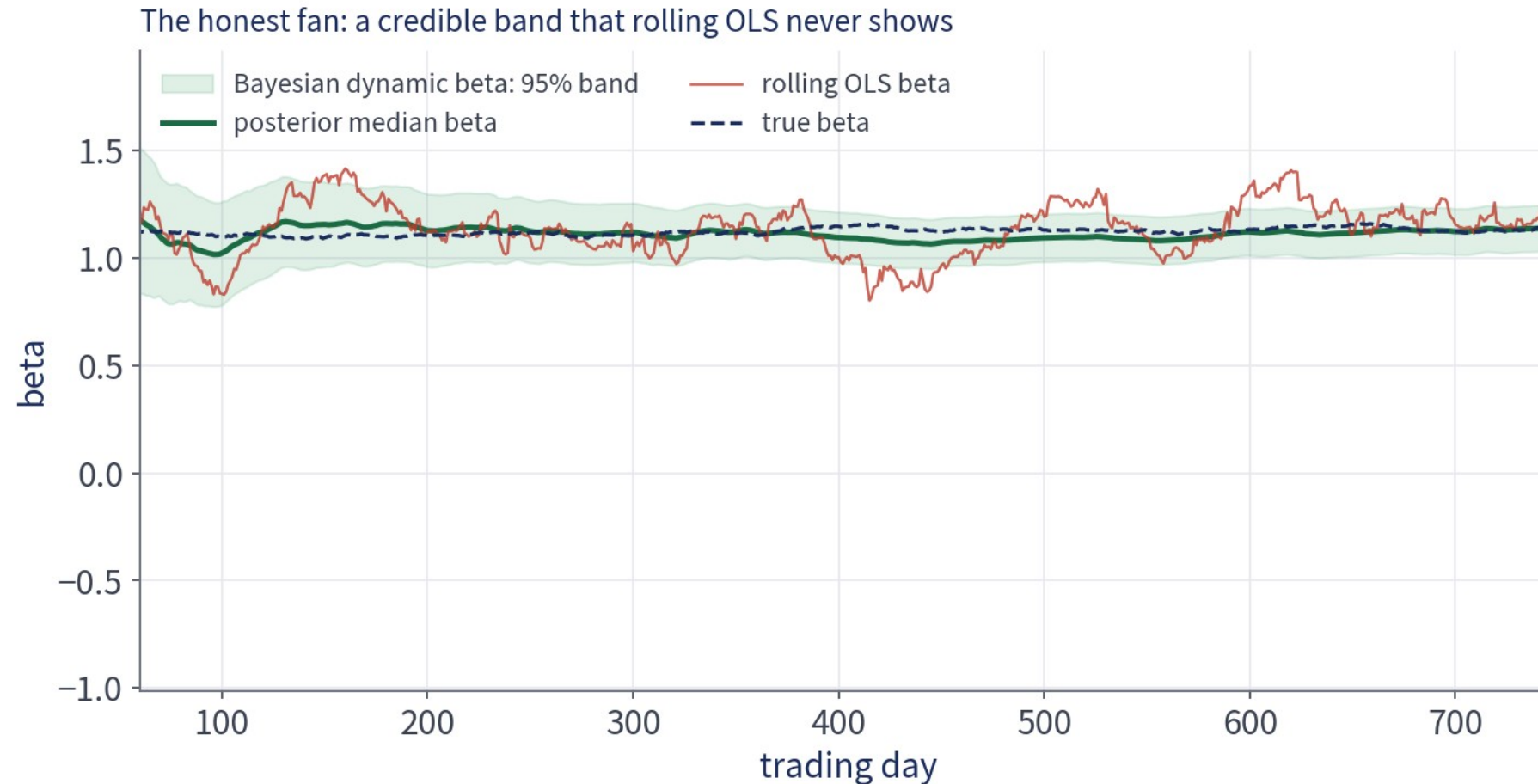
Prior + data = posterior, by hand

$$\begin{aligned}\text{prior } \beta &\sim N(1.0, 0.20^2), & \text{new estimate } y &= 1.4 \pm 0.10 \\ \text{precision} &= \frac{1}{0.20^2} + \frac{1}{0.10^2} = 25 + 100 = 125 \\ \text{posterior var} &= \frac{1}{125} = 0.008 \Rightarrow \text{sd} = \sqrt{0.008} = 0.089 \\ \text{posterior mean} &= \frac{25(1.0) + 100(1.4)}{125} = \frac{165}{125} = \mathbf{1.32} \pm 0.089\end{aligned}$$

The posterior is a precision-weighted average: pulled toward the sharper input, and tighter than both.
That sd is the band.

The Honest Fan for Beta

Repeating that update every day gives the Bayesian dynamic beta: it tracks the true path smoothly AND carries a band. Where the rolling line jumps confidently, the band widens to tell you beta is hard to pin down



What the Overconfident Beta Costs

The band changes real decisions

With the single number

- You hedge as if $\beta = 1.18$ exactly.
- You discount cash flows at one precise rate.
- A wrong beta mis-sizes the hedge and the value.

With the band

- You see when beta is well-pinned or shaky.
- You widen the hedge margin when it is uncertain.
- You stress the valuation across the range.

Skill, or Just Luck?

A great track record - or a short one

A strategy shows a two-year record with a headline Sharpe ratio. Impressive - the kind of number that wins an allocation. We will compute that Sharpe from six months by hand, then compute its error bar, and find the honest interval cannot even rule out zero.

The pitch

- A smooth-looking equity curve.
- A Sharpe that reads as real skill.
- It attracts capital fast.

What we'll do by hand

- Compute the sample Sharpe.
- Compute its standard error.
- Build its 95% interval.

The Track Record: Six Months

Compute the Sharpe by hand

Month	1	2	3	4	5	6
Return (%)	+2.5	-1.0	+3.0	+1.5	-0.5	+3.5

Goal: Sharpe = mean return / standard deviation, then annualize by multiplying by the square root of 12.

The Sample Sharpe

Mean, spread, ratio, annualize

$$\bar{r} = \frac{2.5 - 1.0 + 3.0 + 1.5 - 0.5 + 3.5}{6} = 1.50\%$$

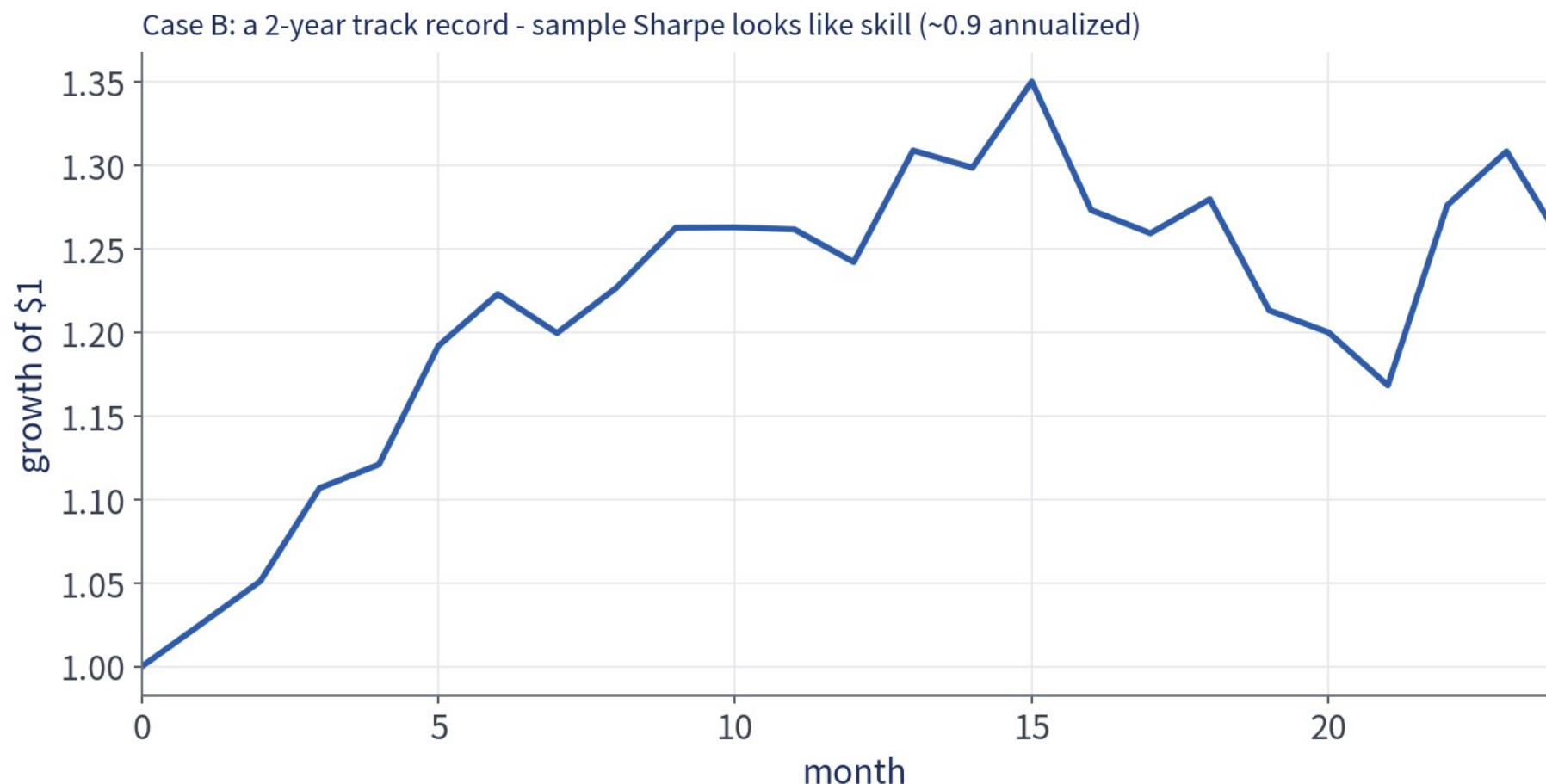
$$s = \sqrt{\frac{\sum (r - \bar{r})^2}{6 - 1}} = \sqrt{\frac{17.5}{5}} = \sqrt{3.5} = 1.87\%$$

$$SR_{\text{month}} = \frac{1.50}{1.87} = 0.80, \quad SR_{\text{ann}} = 0.80 \times \sqrt{12} = \mathbf{2.78}$$

An annualized Sharpe of 2.78 looks spectacular. But it rests on only six data points - so how sure is it?

The Impressive-Looking Track Record

Plotted, those months compound into steady growth with a headline Sharpe. As a point estimate it looks like a clear win - exactly the number that attracts capital. But a point estimate is all it is.



How Uncertain Is a Sharpe?

The formula we'll plug into

$$\widehat{SR} \pm se, \quad se \approx \sqrt{\frac{1 + \frac{1}{2}\widehat{SR}^2}{T}} \quad (\text{short } T \Rightarrow \text{wide band})$$

Reading it

- The Sharpe comes with a standard error.
- It shrinks only with more time, T .
- Six months is very little T .

What to expect

- A wide band around the estimate.
- A headline near three, consistent with zero.
- We compute it exactly next.

The 95% Interval Spans Zero

Plug in and see

$$\text{se}(SR_{\text{month}}) = \sqrt{\frac{1 + \frac{1}{2}SR^2}{T}} = \sqrt{\frac{1 + 0.5(0.80)^2}{6}} = \sqrt{0.220} = 0.469$$

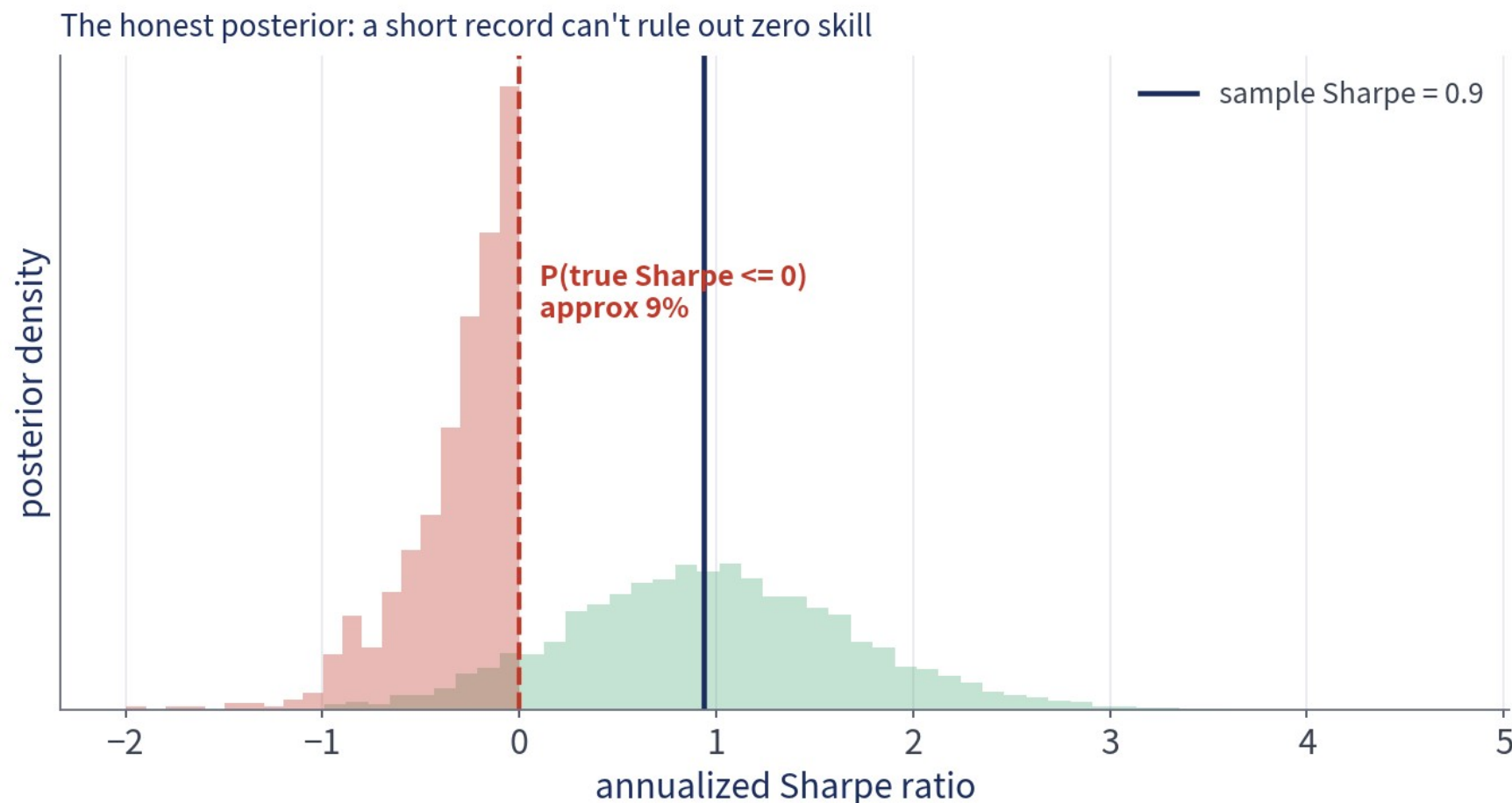
$$\text{se}(SR_{\text{ann}}) = 0.469 \times \sqrt{12} = 1.63$$

$$95\% \text{ CI} = 2.78 \pm 1.96(1.63) = 2.78 \pm 3.19 = (-\mathbf{0.41}, \mathbf{5.96}) \ni 0$$

Even a Sharpe of 2.78 has a 95% interval from -0.41 to 5.96. It includes zero: over six months, skill and luck are indistinguishable.

The Posterior Spans Zero

Resampling the returns gives the same message as the hand calculation: the Sharpe's distribution is wide, and a real share of it sits at or below zero. A confident headline hides a genuine chance of no edge.



What the Overconfident Sharpe Costs

The quant-quake lesson, in miniature

With the single number

- You allocate as if the skill is proven.
- You size the position to a Sharpe near three.
- You may be paying for luck, at full size.

With the interval

- You allocate in proportion to the evidence.
- You start small and add as the record lengthens.
- You demand more data before betting big.

The Common Pattern

Two calculations, one shape

Both cases share a shape you have now computed by hand: a single fitted number drives a real decision while hiding how uncertain it is. Beta sets a hedge; the Sharpe wins an allocation. In each, a Bayesian band turns the deceptive line into an honest fan.

The trap

- A number you act on, taken as exact.
- Uncertainty invisible in the point estimate.
- A decision sized to false precision.

The fix

- Compute the band, not just the point.
- Weight prior and data by precision.
- Size the decision to that honest width.

Where a Confident Number Bites

Beyond beta and the Sharpe, the same trap recurs: factor loadings, volatility and VaR, hedge ratios, duration. Wherever one fitted number drives a decision, the same hand-computable band reveals the risk it hides

Where a Confident Number Bites

The same trap - a number you act on that hides its uncertainty - recurs all over finance.

Market beta

used for: hedging, cost of capital

one line -> a fan

Sharpe / alpha

used for: manager selection

a point -> spans zero

Factor loadings

used for: risk attribution

fixed -> uncertain

VaR / volatility

used for: risk limits

one level -> a band

Hedge ratio

used for: pairs trading

line -> credible band

Duration / DV01

used for: rate hedging

point -> posterior

The pattern, every time

Wherever a single fitted number drives a real decision, a Bayesian state-space model replaces it with a posterior - and the width of that posterior

Reproduce the Hand Calcs, Then Scale Up

Direct Claude Code to check your arithmetic

GOAL Verify both worked examples in code, then extend them to full time series.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - beta:

- * take the 5-day table; compute Cov, Var, beta and print each intermediate number;
- * do the normal-normal update by hand-formula, printing posterior mean and sd;
- * then run a rolling beta and a dynamic-beta band.

CASE B - Sharpe:

- * take the 6-month returns; print mean, sd, monthly and annual Sharpe;
- * compute the SE and the 95% interval;
- * confirm it includes zero, then bootstrap it.

Match every printed number to the slides.

Check against the slides

- $\text{beta} = 1.18$; posterior 1.32 ± 0.09 .
- Sharpe (ann.) = 2.78.
- 95% interval $(-0.41, 5.96)$.
- Both include a value that flips the call.

Deliverable: `part1_practice.py` whose printed numbers match the hand calculations, then the charts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you get beta from the five-day table?

$\text{Cov}(\text{market}, \text{stock}) / \text{Var}(\text{market})$

Q2 In the update, why is the posterior pulled toward the data?

The data had higher precision

Q3 Why can't a 6-month Sharpe of 2.78 prove skill?

Its 95% interval includes zero

All three come from the beta calc, the precision-weighted update, and the Sharpe interval.

Two Calculations to Remember

Part 1 applications in one screen

1

A beta is Cov/Var - and it moves

We computed $\beta = 1.18$ from five days, saw a second window disagree, then updated prior and data into a posterior of 1.32 ± 0.09 . The band is a precision-weighted average, tighter than either input.

2

A short Sharpe can't tell skill from luck

Six months gave an annualized Sharpe of 2.78 - but its 95% interval runs from -0.41 to 5.96. It includes zero, so the impressive number proves nothing yet.

3

Same trap, same fix

Wherever one hand-computed number drives a decision, compute its band too. The width of that band is the risk the point estimate was hiding.

Next - Part 2 practice: a hedge ratio and a VaR, also worked by hand.